Arithmetic Sequences and Series

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Arithmetic sequences

Consider the sequence $7, 11, 15, 19, \ldots$

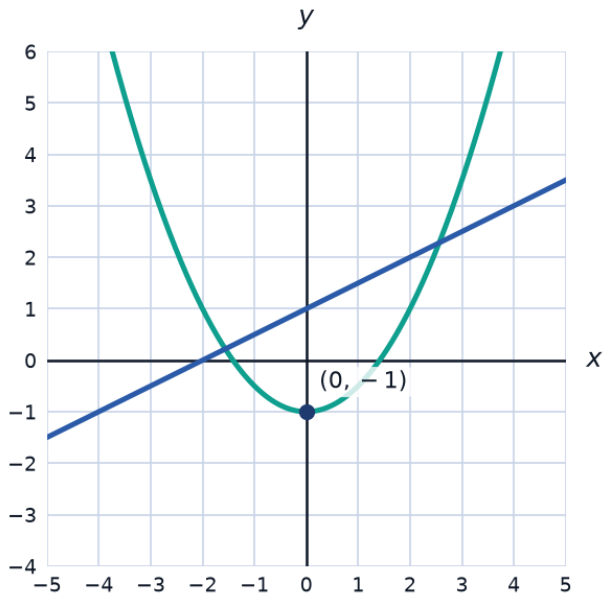
a State the common difference.

b Write a formula for the nth term.

c Find a_{20}.

2 [Foundational] • Target: Arithmetic sequences

In an arithmetic sequence, $a_3 = 10$ and $a_8 = 25$.



a Find the common difference and the first term.

b Write a formula for a_n.

3 [Foundational] • Target: Arithmetic series

Find the sum of the first 30 terms of the series $5 + 9 + 13 + \cdots$

4 [Foundational] • Target: Arithmetic series

Consider the multiples of 6 strictly between 100 and 500.

a How many are there?

b Find their sum.

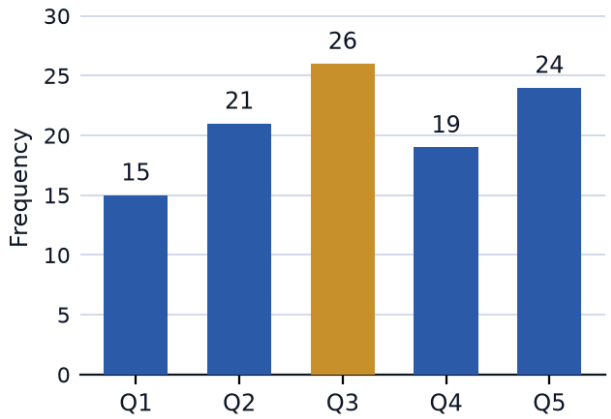
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5 [Foundational] • Target: Arithmetic series

A theatre has 18 seats in the first row, and each subsequent row has 3 more seats than the row before. How many seats are there in total in 25 rows?

6 [Foundational] • Target: Arithmetic means and sequence properties

Insert three arithmetic means between 5 and 25 (i.e., find b_1, b_2, b_3 so that $5, b_1, b_2, b_3, 25$ is arithmetic).



7 [Foundational] • Target: Arithmetic means and sequence properties

Given the arithmetic sequence with $a_1 = -8$ and $d = 3$, find the smallest n such that $a_n > 100$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Arithmetic means and sequence properties

Prove that the sum of the first n positive odd integers $(1 + 3 + 5 + \cdots)$ equals n^2, using the arithmetic series formula.

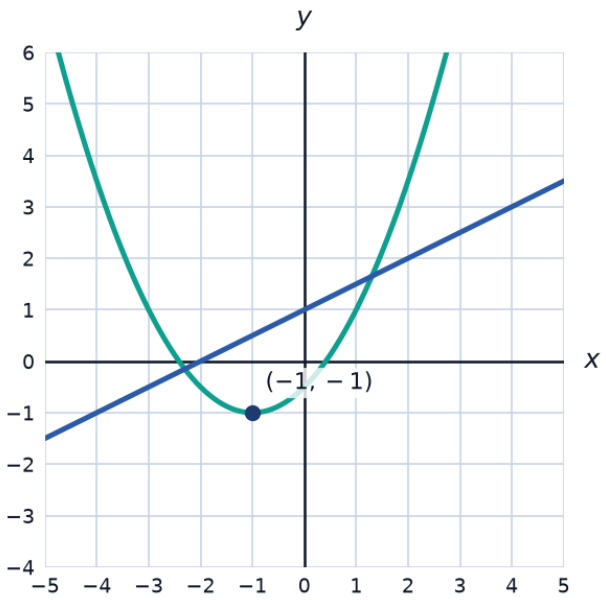
9 [Intermediate] • Target: Applications

A person saves 100 riyals in week 1, and increases savings by 20 riyals each subsequent week. Find the total saved after 20 weeks.

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10 [Intermediate] • Target: Applications

A stack of logs has 20 logs in the bottom row, 19 in the next, decreasing by 1 each row, ending with 1 log in the top row. Find the total number of logs.



11 [Intermediate] • Target: Visualizing an arithmetic sequence

The first 6 terms of the sequence $a_n = 3n - 1$ are plotted below. Use the graph to confirm the sequence is linear in n, and state the common difference from the graph.

12 [Intermediate] • Target: Arithmetic sequences in geometry

The interior angles of a pentagon form an arithmetic sequence. If the smallest angle is $80°$ and the angles sum to $540°$ (the pentagon angle sum), find the common difference.

13 [Intermediate] • Target: Finding a term given the sum

The sum of the first n terms of an arithmetic sequence with $a_1 = 3$ and $d = 4$ is $S_n = 210$. Find n.

14 [Intermediate] • Target: Arithmetic sequences with negative common difference

A sequence begins $50, 44, 38, 32, \ldots$ Find the first negative term.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

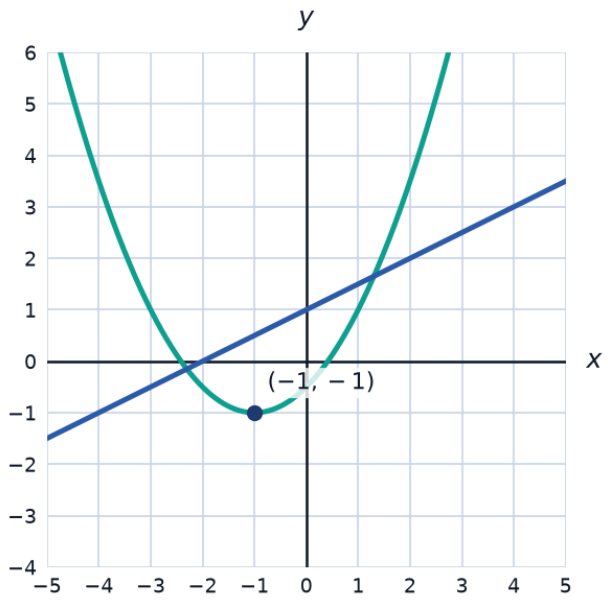
15 [Challenging] • Target: Proving an arithmetic sequence property • Cross-Lesson: 4.02 Ellipses

Prove that for any three consecutive terms a_{n-1}, a_n, a_{n+1} of an arithmetic sequence, the middle term equals the average of its neighbors: $a_n = \frac{a_{n-1} + a_{n+1}}{2}$.

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16 [Challenging] • Target: Arithmetic means and sequence properties • Cross-Lesson: 4.03 Hyperbolas

Consider the arithmetic sequence with first term $a_1 = 5$ and a common difference $d = 3$. What is the formula for the n-th term, a_n? Show your complete algebraic work.



17 [Challenging] • Target: Applications • Cross-Lesson: 5.01 Vectors: Components and Operations

A geometric sequence has its first term $a_1 = 5$ and its common ratio $r = 2$. If the sum of the first n terms of this sequence is $S_n = 315$, find the value of n.

18 [Challenging] • Target: Visualizing an arithmetic sequence • Cross-Lesson: 5.02 The Dot Product

Consider an arithmetic sequence where the 5th term is 19 and the 11th term is 43. If the first term is denoted by a_1 and the common difference by d, find the value of $a_1 + d$.

19 [Challenging] • Target: Arithmetic sequences in geometry • Cross-Lesson: 5.03 Parametric Equations

Consider the arithmetic sequence where the first term is $a_1 = 5$ and the common difference is $d = 3$. What is the 4th term of this sequence, a_4? Show your complete algebraic work.

20 [Challenging] • Target: Finding a term given the sum • Cross-Lesson: 6.01 Polar Coordinates and Conversions

Consider the arithmetic sequence where the first term is $a_1 = 5$ and the common difference is $d = 3$. What is the fourth term of this sequence, a_4? Show your complete algebraic work.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: General Term of an Arithmetic Sequence

State the formula for the n-th term a_n of an arithmetic sequence with first term a_1 and common difference d.

a. $a_n = a_1 + (n-1)d$

b. $a_n = a_1 \cdot d^{n-1}$

c. $a_n = a_1 + nd$

d. $a_n = \frac{a_1 + a_n}{2}$

22 [Foundational] • Target: General Term of an Arithmetic Sequence

Find the 20th term of the arithmetic sequence $4, 7, 10, 13, \ldots$

a. 61

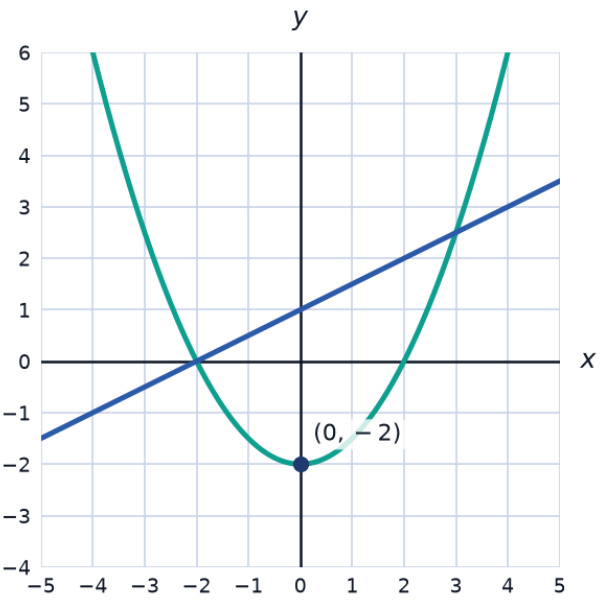
b. 58

c. 64

d. 60

23 [Foundational] • Target: General Term of an Arithmetic Sequence

In an arithmetic sequence, $a_3 = 11$ and $a_8 = 31$. Find the first term a_1 and common difference d.



a. $a_1 = 3, d = 4$

b. $a_1 = 4, d = 3$

c. $a_1 = 2, d = 5$

d. $a_1 = 1, d = 4$

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24 [Foundational] • Target: Sum of an Arithmetic Series

State the two standard formulas for the sum S_n of the first n terms of an arithmetic series.

a. $S_n = \frac{n}{2}(a_1 + a_n)$ and $S_n = \frac{n}{2}[2a_1 + (n-1)d]$

b. $S_n = \frac{a_1(1 - r^n)}{1 - r}$

c. $S_n = n(a_1 + a_n)$

d. $S_n = \frac{a_1 + a_n}{2}$

25 [Foundational] • Target: Sum of an Arithmetic Series

Find the sum of the first 30 terms of the arithmetic series $2 + 5 + 8 + 11 + \ldots$

a. 1365

b. 1350

c. 2730

d. 1410

26 [Foundational] • Target: Sum of an Arithmetic Series

Find the sum of the positive integers from 1 to 100.

a. 5050

b. 5000

c. 5100

d. 10100

27 [Foundational] • Target: Sigma Notation and Number of Terms

Evaluate $\sum_{k=1}^{25} (4k - 1)$.

a. 1275

b. 1300

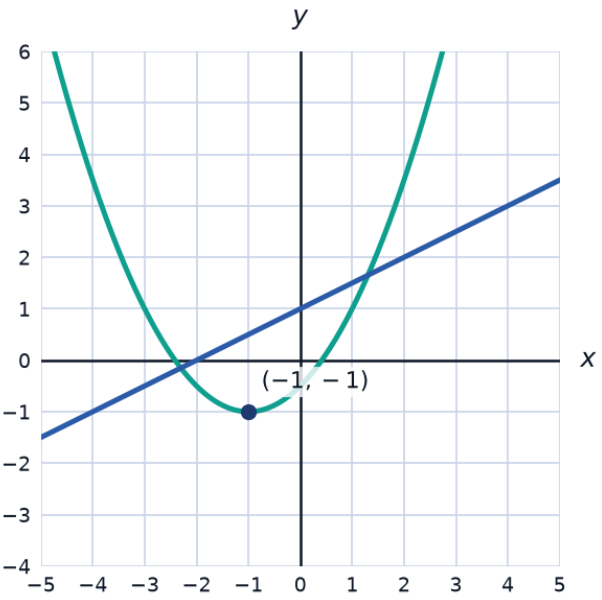
c. 1250

d. 2550

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28 [Foundational] • Target: Sigma Notation and Number of Terms

How many terms are in the finite arithmetic sequence $7, 11, 15, \ldots, 87$?



a. 21

b. 20

c. 22

d. 19

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Sigma Notation and Number of Terms

Evaluate $\sum_{k=5}^{20} 3k$.

a. 600

b. 570

c. 630

d. 300

30 [Intermediate] • Target: Word Problems and Real-World Applications

An auditorium has 20 rows of seats. The first row has 15 seats, and each subsequent row has 2 more seats than the row before it. How many total seats are in the auditorium?

a. 680

b. 660

c. 700

d. 580

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31 [Intermediate] • Target: Word Problems and Real-World Applications

A runner trains by running 3 miles on day 1, and increases her daily distance by 0.5 miles each day. How many total miles will she have run in 14 days?

a. 87.5 miles

b. 91 miles

c. 84 miles

d. 95.5 miles

32 [Intermediate] • Target: Word Problems and Real-World Applications

A clock chimes the number of hours on the hour (1 chime at 1:00, 2 chimes at 2:00, ..., 12 chimes at 12:00). How many total chimes does it make in a 12-hour period?

a. 78

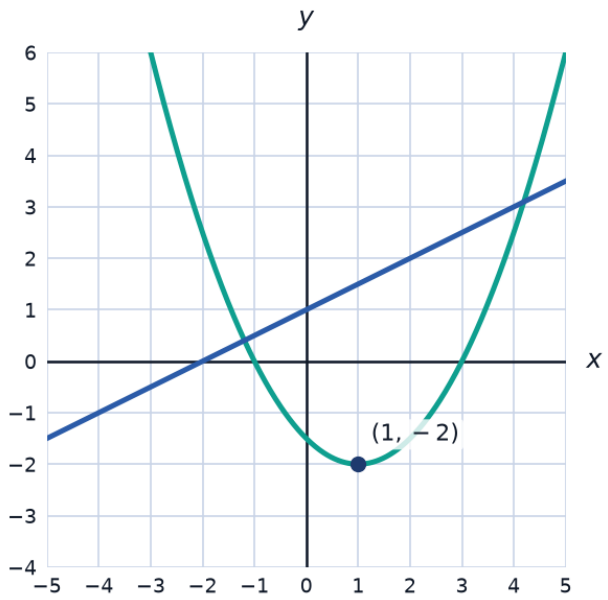
b. 72

c. 84

d. 66

33 [Intermediate] • Target: Word Problems and Real-World Applications

Find the 6th term of the arithmetic sequence with $a_1 = 2$ and common difference $d = -4$.



a. -18

b. -14

c. -26

d. -22

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34 [Intermediate] • Target: Arithmetic sequences

Find the 8th term of the arithmetic sequence with $a_1 = 0$ and common difference $d = -4$.

a. -24

b. -28

c. -36

d. -32

35 [Intermediate] • Target: Arithmetic series

Find the 6th term of the arithmetic sequence with $a_1 = 6$ and common difference $d = -5$.

a. -29

b. -19

c. -14

d. -24

36 [Intermediate] • Target: Arithmetic means and sequence properties

Find the 5th term of the arithmetic sequence with $a_1 = 3$ and common difference $d = -5$.

a. -17

b. -22

c. -12

d. -27

37 [Intermediate] • Target: Applications

Find the 5th term of the arithmetic sequence with $a_1 = 8$ and common difference $d = -5$.

a. -17

b. -22

c. -7

d. -12

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Visualizing an arithmetic sequence • Cross-Lesson: 5.02 The Dot Product

Find the 6th term of the arithmetic sequence with $a_1 = 0$ and common difference $d = -6$.

a. -30

b. -36

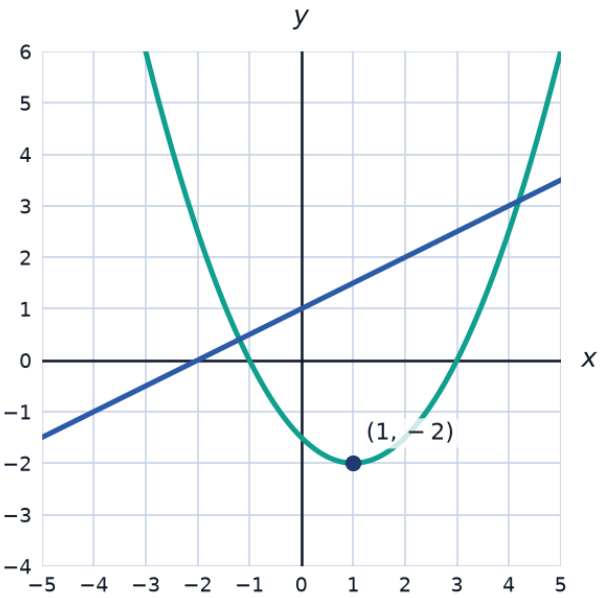
c. -24

d. -42

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39 [Challenging] • Target: Arithmetic sequences in geometry • Cross-Lesson: 5.03 Parametric Equations

Find the 4th term of the arithmetic sequence with $a_1 = 5$ and common difference $d = -5$.



a. -15

b. -10

c. -5

d. -20

40 [Challenging] • Target: Finding a term given the sum • Cross-Lesson: 6.01 Polar Coordinates and Conversions

Find the 8th term of the arithmetic sequence with $a_1 = 3$ and common difference $d = -2$.

a. -13

b. -15

c. -11

d. -9

41 [Challenging] • Target: Arithmetic sequences with negative common difference • Cross-Lesson: 6.02 Polar Equations and Graphs

Find the 6th term of the arithmetic sequence with $a_1 = 1$ and common difference $d = -5$.

a. -29

b. -19

c. -34

d. -24

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42 [Challenging] • Target: Proving an arithmetic sequence property • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

Find the 6th term of the arithmetic sequence with $a_1 = -2$ and common difference $d = -1$.

a. -7

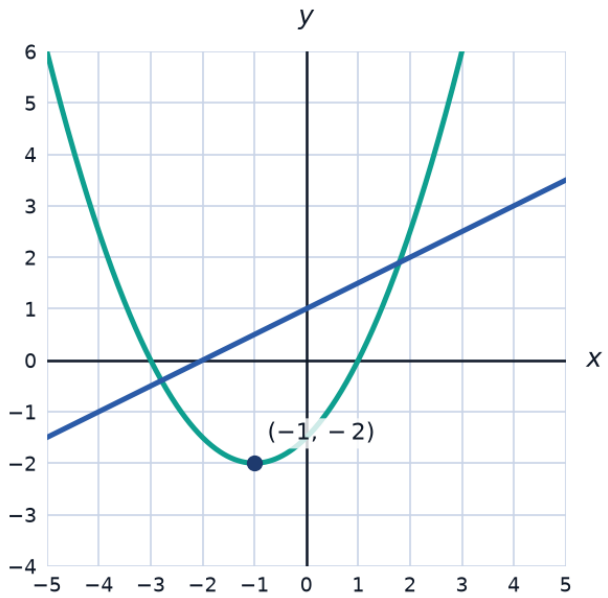
b. -8

c. -6

d. -9

43 [Challenging] • Target: General Term of an Arithmetic Sequence • Cross-Lesson: 7.02 Logarithms and Their Properties

Find the 4th term of the arithmetic sequence with $a_1 = 6$ and common difference $d = -5$.



a. -4

b. -19

c. -14

d. -9

44 [Challenging] • Target: Sum of an Arithmetic Series • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

Find the 8th term of the arithmetic sequence with $a_1 = -2$ and common difference $d = 6$.

a. 40

b. 46

c. 34

d. 52

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45 [Challenging] • Target: Sigma Notation and Number of Terms • Cross-Lesson: 7.04 Applications: Growth and Decay

Find the 5th term of the arithmetic sequence with $a_1 = 6$ and common difference $d = -2$.

a. -4

b. -2

c. 0

d. -6

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